COMMERCIAL ITEM DESCRIPTION

WHEEL, ABRASIVE (Silicon Carbide, Impregnated Rubber, Fine, Square Edge, 2 by 1/4 inches)

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers a silicon carbide abrasive wheel suitable for use on standard dental lathe chucks for fine grinding and polishing.

Salient characteristics:

Material. The wheel shall be composed of a homogeneous distribution of high quality silicon carbide abrasive, impregnated in a resilient rubber bonding agent. The rubberized abrasive wheel shall contain no ferrous metals, shall be non-magnetic, and shall be of uniform texture throughout, without hard or soft spots. The wheel shall have an open structure.

Abrasive. The abrasive shall be silicon carbide meeting the requirements for No. 120 grit as specified in American National Standards Institute (ANSI) B74.12, controlled within the limits for sizing as specified herein.

Bonding agent. The bonding material shall be oil and age-resistant rubber, free from extraneous material and suitable in all respects for the purpose intended.

Design and dimensions. The abrasive wheel shall be flat with a square edge having the following dimensions: 2 ± 1/16 inch in diameter, 1/4 ± 1/32 inch thick, and 1/4 ± 1/64 inch arbor hole. The wheel shall run true when mounted on a spindle.

Life expectancy. The wheel shall resist deterioration and shall be chemically and physically stable. The wheel shall remain flexible and suitable for use intended when stored at 50 to 90 degrees Fahrenheit.

AMSC N/A FSC 6520

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Performance. The wheel shall be suitable for fine grinding and polishing when mounted on standard dental lathe chuck (see note section) on dental grinding and polishing machine (see note section). The wheel shall not glaze, fill-in, or smear when used on soft metals. When use on hard metals, the wheel shall give a clean, smooth finish or high polish, free of waviness, high spots, or discoloration. The wheel shall not shrink or change texture.

Identification marking. Each wheel shall be permanently and legibly labeled with the manufacturer's name or registered trademark, and maximum operating speed of 10,500 revolutions per minute (r.p.m).

Workmanship. Wheels shall be free of surface irregularities or any other defects which detract from its appearance or may impair its serviceability.

Unit. Each (EA). One wheel, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one wheel. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except for the dimensions the inspection level shall be S-1 AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1. For tests, acceptance number shall be zero (0) for all sample sizes.

Examination. Examination shall include but shall not be limited to the following defects:

Defects.

Rubber not free from extraneous material.
Rubberized abrasive wheel is not of uniform texture, and has hard or soft spots.
Abrasive material not as specified.
Abrasive wheel dimensions not as specified herein.
Label on wheel not permanent or legible.

Tests.

Abrasive grain size. Test for abrasive grain size shall be as specified in ANSI B74.12.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-568-5150	WHEEL, ABRASIVE, SILICON CARBIDE, Impregnated Rubber, Fine, Square edge, 2 by 1/4 inches

The following National Stock Numbers are used with the above wheel:

6520-00-538-9150	GRINDING AND POLISHING MACHINE, Dental Laboratory, Bench Mounted, 115 Volt
6520-00-539-0550	CHUCK, GRINDING AND POLISHING MACHINE, Dental Laboratory, Stone, Right

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS

PROJECT NO. 6520-2095

LOCATION: NCR Disk No. 7

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53754	WHEEL, ABRASIVE (SILICON CARBIDE, IMPREGNATED RUBBER)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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